A positive equilibrium at the critical Dirichlet eigenvalue

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Abstract

We construct a connected unbounded uniformly smooth domain in $\mathbb{R}^{21}$ whose bottom Dirichlet eigenvalue is 1, but on which $-\Delta u = u - u^2$ has a bounded positive solution with zero boundary values. Thus equality between the spectral threshold and the derivative of a strict KPP reaction at zero does not force extinction on all uniformly smooth domains. The domain is a slight widening of a slab of width π. Horizontal Hardy energy controls the transverse spectral deficit, while an explicit positive subsolution survives the quadratic reaction loss.

1 The critical existence question

Consider the Dirichlet equilibrium equation

$$-\Delta u = f(u) \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega, \tag{1}$$

where $\Omega \subset \mathbb{R}^d$ is connected and unbounded. We use the spectral normalization

$$\lambda_1(\Omega) = \inf_{0 \neq \phi \in C_c^\infty(\Omega)} \frac{\int_\Omega |\nabla \phi|^2}{\int_\Omega \phi^2}. \tag{2}$$

For a reaction satisfying $f(0) = f(1) = 0$, $f'(0) > 0$, positivity on $(0,1)$, negativity above 1, and strict decrease of $f(s)/s$ on $(0,1]$, the critical case is $\lambda_1(\Omega) = f'(0)$. In the population interpretation, this balances linear growth against diffusive loss.

Berestycki and Graham prove existence when $\lambda_1(\Omega) < f'(0)$ and nonexistence when $\lambda_1(\Omega) > f'(0)$, as well as critical nonexistence on bounded or periodic domains [1, Theorem 1.5]. Their Conjecture 1.7 asks for critical nonexistence in general domains. Uniform boundary regularity is part of that setting: the boundary graph charts have a common radius and uniformly bounded $C^{2,\gamma}$ norms.

Rossi and Tough establish dimension-dependent critical behavior under quantitative conditions on the nonlinear loss [2]. Their inverse-square potential construction involves the Hardy constant [2, Theorem 4.1]. They also obtain nonuniqueness for the homogeneous logistic equation on locally, but not uniformly, analytic domains [2, Remark 4.2(e) and Section 7]. We give an explicit critical existence example with uniform boundary regularity and a homogeneous reaction.

Write $(x, z) \in \mathbb{R}^{20} \times \mathbb{R}$, and define

$$t(x) = 1000 + |x|^2, \qquad b(x) = \frac{50}{t(x)}, \qquad h(x) = \arccos(-b(x)), \qquad \Omega = \{(x,z) : |z| < h(x)\}. \tag{3}$$

To make the global regularity of the reaction explicit, take

$$f(s) = \begin{cases} s - s^2, & 0 \le s \le 1, \\ s^{-1} - 1, & s \ge 1. \end{cases} \tag{4}$$

The two branches agree in value and first derivative at 1; the function and its derivative are bounded, and its derivative is globally Lipschitz. It satisfies all the reaction conditions above.

Theorem 1.1. *For every fixed $0 < \gamma \le 1$, the domain Ω in* (3) *is connected, unbounded and uniformly $C^{2,\gamma}$, and the reaction* (4) *belongs to $C^{1,\gamma}([0,\infty))$. Moreover,*

$$\lambda_1(\Omega) = f'(0) = 1,$$

and (1) *admits a solution $U \in C^2(\Omega) \cap C(\overline{\Omega})$ with $0 < U \le 1$.*

Here λ_1 is precisely the compact-test infimum (2), and the boundary condition holds at every point of $\partial\Omega$. Since $U \le 1$, it solves the classical logistic equation. This single example answers the general critical nonexistence question in the negative. The dimension and constants are convenient choices; no optimal dimension is asserted.

The domain contains the critical slab of width π, but is wider by an amount of order $|x|^{-2}$. The horizontal Hardy inequality pays for the resulting decrease of the vertical eigenvalue. A cosine with a small positive offset then supplies a nonlinear subsolution. Section 2 proves the spectral equality, and Section 3 constructs the equilibrium and recovers its boundary values. All functions and test spaces are real.

2 Geometry and spectrum

Lemma 2.1. *The domain in* (3) *is connected, unbounded and uniformly $C^{2,\gamma}$ for $0 < \gamma \le 1$, and $\lambda_1(\Omega) = 1$.*

Proof. The function b has bounded derivatives of every order and takes values in $(0, 1/20]$. The same is true of the positive-order derivatives of h. Also $h > \pi/2$ and

$$|\nabla h| \le 2|\nabla b| \le \frac{200}{t^{3/2}} < \frac{1}{10}.$$

Thus the two global graphs $z = \pm h(x)$ are uniformly separated and have uniformly bounded $C^{2,\gamma}$ norms on fixed-radius charts. For example, charts of horizontal radius $1/4$ have height variation less than $1/40$; the other graph is more than 3 away vertically. The midplane gives connectedness, and Ω contains $S = \mathbb{R}^{20} \times (-\pi/2, \pi/2)$.

We first bound the transverse spectral deficit. For $0 \le b \le 1/20$, set $\theta = 2\arcsin(b)/\pi$. Since $\arcsin(b) \le 6b/5$ and $\pi > 3$,

$$1 - \left(\frac{\pi}{2\arccos(-b)}\right)^2 = 1 - (1+\theta)^{-2} \le 2\theta \le \frac{8}{5}b. \tag{5}$$

The derivative bound $2/\sqrt{3} < 6/5$ on $[0, 1/2]$ proves the arcsine estimate used here.

For $\psi \in C_c^\infty(\mathbb{R}^{20})$, expanding a square gives

$$0 \le \int_{\mathbb{R}^{20}} \left|\nabla\psi + 9\frac{x}{|x|^2}\psi\right|^2 = \int_{\mathbb{R}^{20}} |\nabla\psi|^2 - 81\int_{\mathbb{R}^{20}} \frac{\psi^2}{|x|^2}. \tag{6}$$

Indeed, $\operatorname{div}(x/|x|^2) = 18/|x|^2$. Integrating first outside a ball of radius ε justifies the computation at the origin: the extra boundary term is $O(\varepsilon^{18})$ and tends to zero.

Let $\phi \in C_c^\infty(\Omega)$ and extend it by zero to $\mathbb{R}^{21}$. Its extension remains smooth. Apply (6) to each horizontal slice, and the interval Dirichlet inequality to each vertical slice. For clarity, the latter follows by writing $p(z) = \cos(\pi z/(2H))$ on $(-H, H)$ and using

$$\int_{-H}^{H} \left(|g'|^2 - \frac{\pi^2}{4H^2} g^2 \right) = \int_{-H}^{H} p^2 \left| (g/p)' \right|^2 \geq 0 \qquad (g \in C_c^\infty(-H, H)).$$

With $r = |x|$, Fubini and (5) therefore give

$$\int_\Omega |\nabla \phi|^2 \geq \int_\Omega \left(1 - \frac{80}{1000 + r^2} + \frac{81}{r^2} \right) \phi^2 \geq \int_\Omega \phi^2. \tag{7}$$

The weighted term is integrable near $r = 0$ in twenty dimensions. This proves $\lambda_1(\Omega) \geq 1$.

For the reverse bound, choose $p_\varepsilon \in C_c^\infty(-\pi/2, \pi/2)$ converging to $\cos z$ in H^1. Such functions are obtained by cutting off within distance 2ε of the endpoints; there $\cos z = O(\varepsilon)$, so the derivative error tends to zero in L^2. For any nonzero $\eta \in C_c^\infty(\mathbb{R}^{20})$, the product $\eta(x/L)p_\varepsilon(z)$ is a compact test in $S \subset \Omega$ and has Rayleigh quotient

$$L^{-2} \frac{\int |\nabla \eta|^2}{\int \eta^2} + \frac{\int |p_\varepsilon'|^2}{\int p_\varepsilon^2}.$$

Letting $L \to \infty$ and then $\varepsilon \to 0$ proves the upper bound. □

3 A positive equilibrium

Lemma 3.1. *The smooth function*

$$v(x, z) = \frac{\cos z}{t(x)} + \frac{50}{t(x)^2} \tag{8}$$

is positive in Ω, vanishes on $\partial\Omega$, is bounded by $21/20000$, and satisfies $-\Delta v < v - v^2$ in Ω.

Proof. Positivity and the boundary values follow from $\cos h(x) = -50/t(x)$ and $h < \pi$. With $r = |x|$ and $t = 1000 + r^2$, direct differentiation yields

$$\Delta_x(t^{-1}) = -40t^{-2} + 8r^2 t^{-3}, \qquad \Delta_x(50t^{-2}) = -4000t^{-3} + 1200r^2 t^{-4}.$$

The first coefficient is negative and at least $-40t^{-2}$. The vertical cosine term cancels in $\Delta v + v$, so

$$\Delta v + v = \Delta_x(t^{-1}) \cos z + \Delta_x(50t^{-2}) + 50t^{-2} \geq \frac{10}{t^2} - \frac{4000}{t^3} \geq \frac{6}{t^2}.$$

On the other hand, $0 < v \leq (21/20)t^{-1}$ and $v^2 \leq 441/(400t^2) < 6/t^2$. □

Proof of Theorem 1.1. The reaction properties follow from (4), and Lemma 2.1 proves the geometric and spectral claims. It remains to construct a classical Dirichlet solution above the function v of Lemma 3.1.

Fix $R > 1$ and put $D_R = \Omega \cap \{|x| < R\}$. This is a bounded Lipschitz domain: the map $(x, y) \mapsto (x, h(x)y)$ sends $B_R^{20} \times (-1, 1)$ smoothly onto it, including its transverse corners. Set $w_0 = v$ and define w_{k+1} by

$$(-\Delta + 1)w_{k+1} = G(w_k) \quad \text{in } D_R, \qquad w_{k+1} - v \in H_0^1(D_R), \qquad G(s) = 2s - s^2. \tag{9}$$

All boundary data here are weak traces. After subtracting v, the bilinear form $\int_{D_R} (\nabla a \cdot \nabla c + ac)$ is the full H_0^1 inner product. Lax–Milgram therefore defines every iterate whose right side is

in L^2 [4, Theorem 4.20]. Weak comparison is obtained by testing a difference with its positive part, which has zero trace when the boundary difference is nonpositive; the Sobolev and trace facts used here hold on bounded Lipschitz domains [3, Section 1.1]. Since G is nondecreasing on $[0,1]$, v is a subsolution, and 1 is a supersolution, induction gives

$$v \le w_k \le w_{k+1} \le 1. \tag{10}$$

The limit in this iteration preserves the nonlinear equation. If $w(F)$ denotes the affine Dirichlet solution with right side F, testing the difference of two equations gives

$$\|w(F) - w(F')\|_{H^1(D_R)} \le \|F - F'\|_{L^2(D_R)}.$$

The monotone limit U_R in (10) satisfies $G(w_k) \to G(U_R)$ in $L^2(D_R)$ by dominated convergence. The displayed stability estimate then identifies U_R with $w(G(U_R))$ in H^1. Thus $U_R - v \in H^1_0(D_R)$ and $-\Delta U_R = U_R - U_R^2$ weakly.

On each fixed interior ball the right side is bounded by $1/4$ in absolute value. Interior regularity first gives uniform $C^{1,1/2}$ bounds and then, by polynomial composition and Schauder estimates, uniform $C^{2,1/2}$ bounds on smaller balls [3, Proposition 2.18 and Corollary 2.16]. Taking $R \to \infty$ along a diagonal subsequence gives a limit $U \in C^2(\Omega)$ solving the equation, with $v \le U \le 1$.

To retain the zero boundary values, write $q(x,z) = \cos z + b(x)$ and work in the two collars $0 < q < 1/4$. There define

$$W(x,z) = \frac{1 - e^{-4q(x,z)}}{1 - e^{-1}}.$$

Since $|\cos z| = |q - b| \le 1/4$ and $\Delta_x b < 0$,

$$|\nabla q|^2 \ge \frac{15}{16}, \qquad \Delta q \le 1, \qquad -\Delta W = \frac{4e^{-4q}}{1 - e^{-1}} \left(4|\nabla q|^2 - \Delta q\right) \ge \frac{11}{e - 1} > \frac{1}{4}.$$

The function of q defining W is concave, equals 0 at $q = 0$ and 1 at $q = 1/4$, and hence $W \ge 4q$ throughout the collar. On the boundary of $D_R \cap \{0 < q < 1/4\}$ the traces satisfy $U_R \le W$: both are zero at $q = 0$, $U_R \le 1 = W$ at $q = 1/4$, and the artificial side $r = R$ has trace $v = q/t \le W$. Each collar component is bounded Lipschitz, so weak comparison with $-\Delta U_R \le 1/4 < -\Delta W$ gives $U_R \le W$ there. This bound is independent of R. Passing to the limit yields $0 < U \le W$ near the original boundary. As $q \to 0$, also $W \to 0$, which proves that U extends continuously by zero to every point of $\partial\Omega$. Finally, $U \le 1$ means $f(U) = U - U^2$, as required. □

AI use. OpenAI Codex was used to develop and check the construction and to prepare the exposition. Independent mathematical review is requested.

References

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